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The Indian Gut Health Playbook

2026-27 DIGITAL EDITION

Acidity, gas, bloating, constipation and IBS - what the Indian evidence says, and what to do next

42 chapters. 6 printable worksheets. Every figure traced to ICMR-NIN, the Indian Society of Gastroenterology, the Rome Foundation, peer-reviewed Indian studies, FSSAI regulations or the CGHS rate list.

Read this first

> This is a working playbook, not a textbook. It is written so that a person with ordinary symptoms can name the pattern, order the right tests, and stop paying for things that do not work.

Every number in this book comes from a named source: ICMR-NIN, the Indian Society of Gastroenterology, the Rome Foundation, peer-reviewed Indian journal studies, the CGHS rate list, FSSAI regulations, or major international trials. Where the evidence is weak, the book says so. Where Indian data differ from Western data, the Indian data win.

How to use it

1. Start with chapter 2. Match your symptoms to a pattern in the decoder table.
2. Read chapter 3 before anything else if you have any red flag on the list.
3. Go to the chapter for your pattern, and follow its tests and its food section.
4. Print chapters 37 to 42. They are designed to be filled in by hand.
5. Give it four weeks before judging any change. Every trial in this field uses four weeks or more.

What this book is not

It is not a diagnosis and it is not a prescription. It does not tell you which antibiotic to take, which dose to use, or when to stop a medicine - those are decisions made by a doctor who has examined you. What it does is make you a better-informed patient: able to describe your pattern precisely, ask for the right test, understand the report, and recognise when a symptom needs attention rather than a tablet.

Gut symptoms are common, and the great majority are not dangerous. That is exactly why the rare dangerous ones get missed: they hide inside a crowd of ordinary complaints. The red flag list in chapter 3 is the most important page in this book.

A note on the numbers

Prevalence figures for functional gut disorders in India range from 7% to 66% depending on the definition and the survey method. That is not a contradiction in the science; it is what happens when you measure a symptom pattern instead of a lab value. Read every percentage as a description of a population, never as a prediction about you.

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PART 1
Name the problem

1. "Gas problem" is four different problems

> Almost nobody has just gas. What you call gas is usually one of four patterns, and each one needs a different fix.

Walk into any Indian pharmacy and ask for something for gas. You will be handed an antacid, a carminative, a probiotic sachet, or all three. That is because the word gas is used for at least four unrelated things: acid coming up, food sitting heavy in the stomach, gas trapped and distending the abdomen, and a bowel that moves too fast or too slow.

The four patterns look different once you write them down honestly:

- **Reflux pattern.** Burning behind the breastbone, sour fluid coming up, worse after a heavy meal or on lying down. The problem is at the valve between stomach and food pipe, not in the bowel.
- **Dyspepsia pattern.** Fullness after eating a normal meal, feeling full after a few bites, pain or burning in the upper middle abdomen. The problem is how the stomach empties and how it senses stretch.
- **Bloating pattern.** The abdomen visibly swells during the day, clothes feel tight by evening, relief after passing wind or stool. Distension is often a reflex of the abdominal wall, not litres of gas.
- **Bowel pattern.** Pain that eases after passing stool, a change in how often or how formed the stool is. This is the irritable bowel pattern, and it is defined by the stool, not by the gas.

Two or three of these commonly travel together. In an Indian gastroenterology outpatient study published in 2025, about half of all attendees had at least one functional gut disorder, and 40% had two or more overlapping. So overlap is normal, not a sign that something rare is going on.

Why this matters: the treatment for each pattern is different. An antacid helps reflux and does little for constipation. A fibre supplement helps constipation and can make reflux-related bloating worse for the first two weeks. A probiotic may help one specific pattern and is a waste of money for another. Naming the pattern is 80% of the work.

2. The symptom decoder: twelve patterns

> Match your own words to the pattern, then read the chapter for that pattern. Guess nothing.

What you say	Most likely pattern	Read
Burning in the chest after food, sour taste in the mouth	Reflux	Chapter 5
Food feels stuck high up, or a lump in the throat	Globus or oesophageal sensitivity	Chapter 5
Upper-middle stomach pain or burning, worse when hungry	Dyspepsia, possibly ulcer	Chapters 4 and 6
Full after five bites, heavy for hours	Functional dyspepsia, fullness type	Chapter 4
Belly swells by evening, tight clothes, no pain	Functional bloating	Chapter 29
Loud gurgling, wind, socially embarrassing	Belching or gas transit, often benign	Chapter 29
Hard stool, straining, twice a week	Functional constipation	Chapter 28
Pain then loose stool, urgency, three times before noon	IBS, diarrhoea type	Chapter 27
Alternating hard and loose stools with pain	IBS, mixed type	Chapter 27
Mucus in stool, feeling of incomplete emptying	IBS pattern, common in Indian data	Chapter 27
Loose stools only after milk or curd	Lactose malabsorption	Chapter 14
Symptoms began within weeks of a stomach infection or antibiotics	Post-infectious IBS	Chapter 32

Two cautions. First, this table maps symptoms to patterns, not to diagnoses; a doctor still has to confirm. Second, if any red flag from chapter 3 is present, skip the decoder and get seen.

3. Nine red flags that end self-treatment

> These nine findings change the plan from "manage it" to "investigate it". None of them is common. All of them are worth one appointment.

1. **Difficulty swallowing**, or food sticking on the way down, at any age.
2. **Vomiting blood**, or vomit that looks like coffee grounds.
3. **Black tarry stools**, or fresh blood mixed in stool.
4. **Unintentional weight loss** - roughly 5% or more of body weight in a few months without trying.
5. **Anaemia or low ferritin** found on a blood test, especially with gut symptoms.
6. **A new, persistent change in bowel habit after age 45** that lasts more than four to six weeks.
7. **Persistent vomiting**, or vomiting that stops you eating, for more than two weeks.
8. **A first-degree relative with stomach or oesophageal cancer**, or with colorectal cancer before 50.
9. **A lump you can feel** in the abdomen, or pain that wakes you at night.

Guidelines from the American Society for Gastrointestinal Endoscopy put age 50 as the threshold where new indigestion alone justifies a look inside, even without alarm features. The British Society of Gastroenterology goes further for people from regions where stomach cancer is common: it recommends urgent evaluation from age 40, which includes India. NICE's rule is the simplest to remember - difficulty swallowing, or age 55 and over with weight loss, means an endoscopy within two

weeks.

Notice what is not on the list: bloating, belching, mucus, a normal appetite, symptoms that have not changed in years, or pain that improves after passing stool. Those are common and rarely sinister. The list above is the one that matters.

4. Rome IV, translated: what your doctor is ticking

> Rome IV is a checklist, not a test. It exists so that a symptom pattern can be named without cutting anything open.

The Rome Foundation publishes symptom-based definitions for disorders where no structural damage is found. They are written for research, so the language is clumsy. Here is the plain version of the four that matter to you.

Irritable bowel syndrome

Recurrent abdominal pain, on average at least one day a week in the last three months, plus at least two of: it is related to passing stool, it comes with a change in how often you go, it comes with a change in how the stool looks. Symptoms must have started at least six months before the diagnosis. Pain is compulsory; painless constipation or painless loose stools is not IBS.

Functional dyspepsia

One or more of: bothersome fullness after meals, bothersome early fullness, bothersome pain in the upper middle abdomen, bothersome burning there - with no structural disease found to explain it. It splits into a meal-related type (fullness and early satiety on at least three days a week) and a pain-predominant type.

Functional constipation

Two or more of: straining on more than a quarter of visits to the toilet, hard or lumpy stool on more than a quarter, a feeling of incomplete emptying, a feeling of blockage, needing manual help, fewer than three spontaneous stools a week - with loose stools rarely present unless laxatives were used.

Functional bloating

Recurrent bloating or visible distension that is not part of another disorder and is not explained by a structural cause. It sounds vague because it is: bloating is the least understood of the four, and the one where the abdominal wall's reflex matters more than the gas inside.

One honest note on numbers. The Rome Foundation's worldwide survey found 40% of adults met criteria for at least one of these disorders when asked online, but the same survey's face-to-face arm in India found only 7.2%. In an Indian college sample the figure was 26.9%; in a 2025 house-to-house survey in Ernakulam district, Kerala, it was 66.4%. The range is that wide because the definition and the asking method change the answer. Do not read any single prevalence number as your personal risk.

5. Reflux: heartburn, globus, and the cough that is not asthma

> Reflux is common in India - pooled prevalence about 15.6% - and in most Indian patients the endoscopy looks normal.

A 2021 meta-analysis of nine Indian population studies covering 20,614 people put the pooled prevalence of gastro-oesophageal reflux disease at 15.6%, with individual studies ranging from 5% to 28.5%. The Indian Society of Gastroenterology's own position statement gives a range of 7.6% to 30%, under 10% in most population studies, and notes that fewer than 10% of Indian reflux patients have visible erosive damage on endoscopy. That last number is the reassuring one: for most people this is a symptom problem, not a wound.

What reflux is

A ring of muscle at the junction of food pipe and stomach should close between swallows. When it relaxes at the wrong time, stomach contents move up. Acid in the food pipe produces burning. That is the whole mechanism for the common form.

The 2020 API-ISG consensus guideline is unusually practical: diagnose on the classic symptoms - heartburn and sour regurgitation - and start empirical treatment. Reserve invasive tests for alarm symptoms and for people who do not respond.

Three impostors

- **Globus.** A lump in the throat that is not there, felt between meals, with no difficulty swallowing. It is a sensory problem in the throat, not a growth. It is common, it is harmless, and it responds poorly to acid suppression.
- **Reflux hypersensitivity.** Normal acid exposure, normal endoscopy, but the food pipe reacts to normal amounts of acid. Diagnosed only with pH testing. Responds to different drugs than classic reflux.
- **Functional heartburn.** Burning behind the breastbone with no reflux and no response to acid suppression at all. Chasing it with stronger acid blockers is the wrong direction.

What actually helps

Weight loss where weight is a factor, not lying down within three hours of the main meal, raising the head of the bed by 15 to 20 cm (raise the frame, not the pillow), smaller meals, and stopping smoking. In Indian data, non-vegetarian food, fried food, aerated drinks and tea or coffee were associated with reflux symptoms, and higher body mass index with symptoms in employed urban groups. None of these is a magic switch; each moves the odds a little.

One caution that saves money: acid suppression has no proven benefit for the classic throat-clearing, hoarseness and globus trio. A trial of a proton pump inhibitor for those symptoms is one of the most common wasted prescriptions in India, and the AGA explicitly lists it as a reason to stop the drug.

PART 2
The tests, in the right order

6. Test for H. pylori first, because 58% of Indians carry it

> In a country where more than half of adults carry this bacterium, guessing costs more than testing.

Helicobacter pylori is a spiral bacterium that lives in the stomach lining. It causes chronic gastritis, most duodenal ulcers, some stomach ulcers, and it is classified as a class I carcinogen for stomach cancer. A 2025 meta-analysis of 52 Indian studies across 15 states found a pooled prevalence of 54% among people with gut disease and 61% among people with no diagnosed gut condition - a combined figure of 58%. State-level estimates ranged from about 9% in Gujarat to about 69% in Rajasthan.

That single number decides the strategy. When prevalence in a population is above roughly 20%, guidelines recommend test-and-treat rather than empirical acid suppression for new indigestion in people under 50 with no alarm features. India is far above that threshold.

There is a second reason. A 2025 pan-Indian endoscopy audit of 1,701 patients across 171 centres found H. pylori in 48% of the 905 patients who were tested. A separate prospective study of 500 dyspeptic patients in eastern Bihar found the same shape of result: 55.6% positive overall, and 38.2% positive even among those whose stomach lining looked normal on endoscopy. You cannot see this infection. You have to test for it.

Why eradication is worth it

Eradication cures most duodenal ulcers, reduces the risk of ulcer bleeding, and - per the British Society of Gastroenterology's 2022 dyspepsia guideline - is effective treatment for functional dyspepsia in a meaningful share of patients, not only for ulcers. It also lowers stomach cancer risk. The catch is that India has rising antibiotic resistance, which is why the choice of regimen and the confirmation of cure matter.

7. Which H. pylori test to ask for

> Two tests are useful, one is a trap, and timing around your acid medicine decides whether the answer is real.

Test	What it tells you	Watch out for
Urea breath test	Active infection, right now	Needs a 2-week PPI washout; not for children under 12 in most labs
Stool antigen test	Active infection, right now	Also needs a PPI washout; cheap and widely available
Rapid urease test	Active infection, but only during endoscopy	Requires a biopsy; adds a few hundred rupees to the scope
Blood antibody (IgG)	Only that you were exposed at some point	Cannot tell current from past infection - do not treat on this

NICE states the rule plainly: stop the proton pump inhibitor for two weeks before testing. Antibiotics and bismuth need a four-week gap. Test while on a PPI and you can get a false negative, then spend a year

treating the wrong thing.

If you are being tested for cure after treatment, wait at least four weeks after finishing antibiotics, and stop the PPI for two weeks first. Guidelines only insist on a cure check for people at higher cancer risk or whose symptoms persist, but given Indian resistance rates, asking for one is reasonable.

8. Endoscopy: what 1,701 Indian scopes actually found

> The commonest findings are inflammation and reflux damage, and most of them are managed with tablets, not surgery.

A 2025 multicentre audit collected data from 1,701 patients scoped at 171 Indian centres between April 2024 and March 2025. The reasons for scoping were dyspepsia (49%) and reflux symptoms (31%). The findings were unglamorous: non-erosive gastritis in 33.9%, erosive gastritis in 25.6%, reflux oesophagitis in 17.3%. Most reflux damage was mild to moderate. The antrum, the lower part of the stomach, was the affected site in 73.8% of gastritis cases.

That distribution is useful for expectations. If you are scoped for ordinary indigestion, the most likely outcome is a report that says gastritis, with or without H. pylori, and a plan of tablets. Cancer is rare, and the point of the test is to prove it is not there.

What the test feels like

A flexible tube through the mouth, about five to ten minutes, usually with throat numbing and optional sedation. You cannot eat for six hours before. You will need someone to take you home if you are sedated. It is uncomfortable, not painful, and it is over quickly.

Who should ask for one

Anyone with a red flag from chapter 3. Anyone over 50 with new indigestion. Anyone under 50 whose symptoms persist or recur after H. pylori treatment and a proper trial of acid suppression. Anyone with persistent vomiting, or with anaemia and gut symptoms. And anyone from a family with stomach cancer, from age 40.

9. What each test costs in India

> The government's own CGHS rate list gives a fair floor price. Private hospitals charge two to four times more.

Test	CGHS rate, non-NABH	CGHS rate, NABH
Upper GI endoscopy (gastroscopy), with or without biopsy	Rs. 3,400	Rs. 4,000
Lower GI endoscopy (colonoscopy or sigmoidoscopy)	Rs. 5,100	Rs. 6,000
H. pylori breath test or rapid urease test	Rs. 2,457	Rs. 2,890

These are the rates from the CGHS office memorandum dated 3 October 2025, effective 13 October 2025, for metro (Tier X) cities. Tier Y cities are about 10% lower and Tier Z about 20% lower. CGHS rates apply to CGHS beneficiaries at empanelled hospitals; a self-paying patient at a private hospital is charged the hospital's own rate, which is typically higher.

Practical reading: if you are quoted Rs. 12,000 for a diagnostic gastroscopy at a corporate hospital, that is the market, not a scam. If you are quoted Rs. 4,000 at a district charitable hospital, that is also plausible. Ask three places and ask specifically whether the biopsy and the pathologist's report are included - that is where surprise bills appear.

Check your insurance before you book. Day-care endoscopy is usually covered under a standard indemnity policy after the waiting period, but pre-authorisation rules differ by insurer, and a diagnostic scope is sometimes treated differently from a therapeutic one. One phone call to the insurer's helpline before admission saves an argument later.

10. "The report is normal but I still have symptoms"

> A normal endoscopy is information, not a dismissal. It usually means the problem is function, not damage.

This is the single most common frustration in Indian gastroenterology clinics. You paid for a scope, the report says normal study, and you still cannot eat dinner without bloating. Nothing has gone wrong. The test was designed to find structural disease - ulcers, erosions, tumours, coeliac damage - and it did not find any. That is the good outcome.

What a normal scope rules out: ulcer, erosive oesophagitis, most cancers, visible inflammation. What it does not rule out: functional dyspepsia, IBS, functional bloating, lactose malabsorption, small intestinal bacterial overgrowth, coeliac disease (needs duodenal biopsies, not just a look), or a food intolerance.

So the next step is not a stronger tablet. It is a symptom diary, a proper H. pylori test if it was skipped, a coeliac blood test if diarrhoea or anaemia is part of the picture, and a structured food trial. Chapters 17 and 27 walk through this.

PART 3
Food, the Indian way

11. Fibre: the 30-gram target on an Indian plate

> ICMR-NIN recommends about 15 g of fibre per 1,000 kcal, which is 30 g on a 2,000 kcal day. Most urban Indian plates carry 12 to 18 g.

The 2020 ICMR-NIN nutrient requirements report set fibre against energy intake for the first time: about 15 g per 1,000 kcal, or 30 g for a 2,000 kcal day, labelled a safe intake rather than a requirement. The 2024 Dietary Guidelines for Indians restates it as 25 to 30 g a day and adds two structural targets that matter more than the gram count: at least half of your cereals should be whole grain, and millets can be 30% to 40% of your cereal intake.

The gap is not a mystery. The same 2024 guidelines note that cereals supply 50% to 70% of daily energy against a recommended ceiling of 45%, while pulses, eggs and meat supply 6% to 9% against a recommended 14% to 15%. In plain terms: too much refined grain, too few pulses.

What 30 grams looks like

Food (100 g raw)	Fibre
Pulses, whole and split (rajma, chana, moong, urad, toor)	12 g
Nuts	9 g
Millets (ragi, jowar, bajra)	7 g
Cereals, whole wheat atta and rice	6 g
Green leafy vegetables	2 g
Other vegetables, roots and tubers, fruits	2 g
Milk, curd, egg, meat, fish, oil	0 g

Figures are from the ICMR-NIN 2024 Dietary Guidelines food composition table. Two consequences follow immediately. First, doubling your dal portion does more for fibre than switching from wheat to any millet. Second, adding a salad to a refined-flour diet does almost nothing - vegetables are 2 g per 100 g, so a 150 g salad gives you 3 g.

The ICMR-NIN plate, in weights

For a 2,000 kcal day, the My Plate for the Day model suggests 250 g of cereals including millets, 85 g of pulses, 300 ml of milk or curd, 400 g of vegetables (of which 100 g green leafy), 100 g of fruit, 35 g of nuts and seeds, and 27 g of fats and oils. It also caps sugar at 25 to 30 g a day. If you eat roughly that, you clear 30 g of fibre without buying anything in a jar.

Increase slowly

Going from 12 g to 30 g overnight produces exactly the bloating you are trying to cure. Add about 5 g a week, drink more water, and give it four weeks - the same four weeks that the trials use. Fibre also increases flatulence in the short term; that is documented in the meta-analyses and it usually settles.

12. Isabgol: the one supplement with a real trial behind it

> Psyllium husk is the best-evidenced fibre supplement for both constipation and IBS. Dose above 10 g a day, for at least four weeks, with plenty of water.

Isabgol is psyllium husk, sold in every Indian town for a few hundred rupees a pack. Unlike most of the shelf, it has good trial data.

A randomised trial in 275 primary-care patients with IBS compared 10 g a day of psyllium, 10 g of wheat bran, and a placebo for 12 weeks. Psyllium produced adequate relief in 57% of patients in the first month against 35% on placebo, and cut symptom severity by 90 points against 49 on placebo. Bran did no better than placebo overall, and its commonest side effect was worsening symptoms - a third of the bran group dropped out early.

A 2022 meta-analysis of 16 randomised trials in chronic constipation (1,251 people) found fibre beat control overall, with psyllium the standout: response rate ratio 1.82. The analysis also pinned the dose: more than 10 g a day worked, 10 g or less did not, and four weeks or more was needed for stool frequency to improve. A 2026 meta-analysis of 30 IBS trials (1,904 people) again found psyllium specifically effective, with a risk ratio of 1.53.

How to take it

- Start at 3 to 5 g a day, in a full glass of water, and build to 10 to 15 g a day over two to three weeks.
- Take it with or after a meal, never dry, and follow with another glass of water.
- Give it four weeks before judging. Two weeks is too early.
- Expect more wind in the first fortnight. If bloating is severe, halve the dose and go slower.
- Stop and call a doctor if you develop severe pain, vomiting, or cannot pass stool at all - a bulking fibre in a blocked bowel makes things worse.

One honest limitation: isabgol helps stool form and frequency. It is much weaker for pain and for visible distension. If your main complaint is the swelling, chapter 29 is the relevant one.

13. FODMAP, translated to an Indian kitchen

> The low-FODMAP diet works, but the Indian version is a shortlist of four ingredients, not a foreign rulebook.

FODMAPs are short-chain carbohydrates that ferment in the bowel and draw water into it. In people with a sensitive gut, they produce gas, distension and pain. Reducing them helps a share of IBS patients - a network meta-analysis ranks the diet above other dietary approaches, and a 2025 review of six Asian trials found it beat conventional dietary advice.

The problem is translation. A strict Western low-FODMAP protocol asks you to weigh foods against a database. Indian cooking does not work that way, and much of the Indian FODMAP data is missing. The practical answer comes from the Mumbai-Monash collaboration, which analysed 295 Indian recipes: the main contributors were onion, lentils and wheat products. Of the lentil-based dishes, 88% were high in FODMAPs. Around 40% of non-vegetarian dishes were high too, because of onion, garlic and curd in the masala.

So the Indian shortlist is four things: onion, garlic, dal in quantity, and refined wheat. Add milk if you react to it, which is a separate question covered in the next chapter.

What a two-week trial looks like

1. Keep rice, and keep your normal portions of it. Rice is low in FODMAPs and removing it destabilises everything else.
2. Cut onion and garlic, including pastes and powders. Use garlic-infused oil (the flavour compounds are fat-soluble, the FODMAPs are not), asafoetida, ginger, curry leaves, and the white part of spring onion.
3. Halve the dal for two weeks, or swap to sprouted whole moong, and keep the rest of the meal the same.
4. Swap refined wheat roti for rice, rice roti, jowar or bajra for two weeks - not forever.
5. Keep a daily note of pain, wind and belly girth. Judge at day 14, not day 3.
6. Then reintroduce one item every three days. You are looking for your own threshold, not for a permanent ban.

Two warnings from the literature. Long-term strict restriction risks nutrient gaps and has been associated with lower gut microbial diversity; that is why the diet is designed in three phases and not as a lifestyle. And an Indian survey of 400 healthy people and 204 IBS patients found that self-reported intolerances varied enormously between individuals - pulses bothered 31.9% of IBS patients but only 12% of healthy people; garlic bothered 13.2% of patients and 0.3% of healthy people; no healthy person reported wheat sensitivity. The conclusion the authors drew is the one to keep: the individual, not the FODMAP table, should decide the elimination list.

14. Milk: 60-74% malabsorb, 23% react

> Most Indians digest lactose poorly on a lab test. Far fewer actually get symptoms from a normal glass of milk. The difference decides whether you should give up curd.

In a study of healthy volunteers, lactose malabsorption was found in 88.2% of southern Indians and 66.2% of northern Indians on a tolerance test, and 78.9% versus 57.1% on a breath test. Symptoms after a lactose load appeared in 47.4% of the southern group and 15.6% of the northern group. A 2015 multi-centre study led from SGPGI Lucknow put malabsorption at over 74%, and press coverage of it settled on "three out of four Indians".

Now the twist. In an Indian survey of 400 healthy adults and 204 IBS patients, only 23% of healthy people and 29.9% of IBS patients said milk caused them trouble. The authors pointed out the contradiction directly: a population labelled lactase-deficient still drinks close to 80 million tonnes of milk a year and mostly feels fine.

Both facts are true. A lab load of 25 to 50 g of lactose in a fasting person is not the same as 100 ml of milk with a meal. Lactose in a mixed meal empties from the stomach slowly, and curd and paneer carry less lactose than milk because fermentation and straining remove some of it.

The practical test

Do not eliminate dairy on a guess. Run a one-week trial: keep everything else constant, take your usual milk and curd out, and substitute a calcium source - ragi, sesame, almonds, tofu, or a fortified plant milk.

If nothing changes, dairy was never the problem and you have protected your calcium intake. If wind, urgency and loose stools improve within a week and return within a day of reintroducing milk, you have your answer.

If you do react, the order to try is: curd and buttermilk first, then paneer, then small quantities of milk with meals, then lactose-free milk, and only then full avoidance. Curd is usually tolerated because the bacterial cultures have already done part of the work. Removing dairy permanently without a calcium plan is a real cost - ICMR-NIN recommends 300 ml of milk or curd a day, and it is the main calcium source in most Indian vegetarian diets.

15. Fermented foods: what the Stanford trial showed

> Ten weeks of fermented food raised gut bacterial diversity and lowered inflammatory markers. Fibre alone, at the same effort, did not.

In 2021 a Stanford team randomised 36 healthy adults to either a high-fermented-food diet or a high-fibre diet for 10 weeks. The fermented group went from about 0.4 servings a day to 6.3, eating yogurt, kefir, fermented cottage cheese, kombucha, vegetable brine drinks and fermented vegetables. Their gut bacterial diversity rose steadily, and 19 inflammatory proteins in the blood fell. The fibre group raised intake from 21.5 g to 45.1 g a day; diversity did not change, and none of the 19 inflammatory proteins fell.

The finding that surprised the researchers was the mechanism. Fewer than 6% of the new bacterial strains matched strains found in the fermented foods. The microbes were not colonising from the food; the food was changing the environment so that residents already present could flourish. It also explains why the fibre arm failed: people who start with low diversity often no longer carry the bacteria that digest fibre well.

Two limits to keep in view. The trial had 18 people per arm, no true control group, and ran ten weeks. It measured diversity and inflammatory markers, not whether anyone's bloating improved. Diversity is a proxy, not a symptom score.

What this means for an Indian kitchen

You already have the cheapest fermented foods in the world. Curd and buttermilk, idli and dosa batter, dhokla, kanji, fermented rice water, pickles in oil, and lassi are all part of the tradition. A realistic target, borrowed from the trial's ramp-up, is one fermented serving with two meals a day and one as a drink - roughly four to six servings, not six plus supplements.

Practical cautions. Curd bought cold-chain and stored badly is not a probiotic. Homemade curd set overnight in Indian summer is a reliable culture; a tub that has sat warm for a day is not. And if you have histamine intolerance or a sensitive gut, fermented foods can trigger flushing and wind - start with one serving and build.

16. Probiotics: read the label before you pay

> Strain matters, dose matters, and Indian labelling rules let you check both in ten seconds.

Probiotic effects are strain-specific. A product that helps one condition may do nothing for another, and a capsule that lists only "Lactobacillus acidophilus" without a strain code cannot be matched to any trial. FSSAI's rules for probiotic foods are unusually helpful here, because they force disclosure.

Under the Food Safety and Standards (Health Supplements, Nutraceuticals, Food for Special Dietary Use, Food for Special Medical Purpose, Functional Food and Novel Food) Regulations, 2016, a probiotic food must carry, on the pack: the genus and species with a strain designation or culture collection number, the viable numbers at the end of shelf life, the recommended serving size that delivers the effective dose, and storage conditions plus a use-by time after opening. The minimum is 10 to the power 8 colony-forming units in the recommended daily serving. The label must also say "NOT FOR MEDICINAL USE", must not claim to treat or cure a disease, and must not substitute for a varied diet.

The ten-second check

1. Find the strain code - a letter-and-number tag after the species name. No code, no purchase.
2. Find the CFU count and check whether it is "at the end of shelf life" or "at manufacture". Only the first is meaningful.
3. Check the storage line. If it needs refrigeration and the shop stores it on a shelf, walk away.
4. Check the format. Tablets, capsules and drops need prior FSSAI approval; powders and fermented foods do not.
5. Check the claim. "Supports digestive health" is allowed with substantiation; "cures IBS" is not allowed at all.

Where the evidence is strongest: specific strains for antibiotic-associated diarrhoea, for some IBS symptom scores, and for a few other defined conditions. Where it is weakest: generic "gut health" for a person with no specific diagnosis. A 2025 Indian market analysis noted that most consumers cannot tell whether a product contains a meaningful dose of what it claims - which is exactly why the label check matters more than the brand.

17. The seven-day trigger diary

> Seven days of honest notes beats a year of guessing, and it is the single most useful thing you can bring to an appointment.

Memory is a bad instrument for food reactions, because the meal you blame is often not the meal that caused the symptom - fermentation takes hours, and transit takes a day. A written log fixes that.

Log five things, three times a day, for seven days. Nothing more; a complicated diary gets abandoned by day two.

1. **Time and what you ate**, in ordinary words. "Two rotis, dal, bhindi, curd" is enough.
2. **Worst symptom and its score** from 0 to 10.
3. **Bowel movement**: time, Bristol type (chapter 38), and whether you strained.
4. **Belly girth** in centimetres, measured at the navel, once in the morning before eating and once at night. This is the only reliable way to separate real distension from a feeling of fullness.
5. **One line on the day**: sleep, stress, travel, a missed meal, a late night.

Then read it for patterns, not for culprits. Look for a symptom that appears 6 to 24 hours after a specific food on at least three of seven days. Look for a symptom that appears only on work days. Look for a girth difference of more than 3 cm between morning and night - that is measurable distension, and it belongs in a doctor's notes. And look for the boring finding that shows up in most diaries: symptoms track meal timing and stress better than they track any ingredient.

Bring the seven pages to your appointment. A gastroenterologist can work with a diary in five minutes; without one, the consultation becomes a guess about which of twenty foods to eliminate.

18. Spices, tea, coffee, soda: what Indian data links

> The Indian evidence points at tea and coffee, aerated drinks, fried and non-vegetarian food, and tobacco - not at chilli.

This is where folklore and data part company. In a rural Uttar Pradesh survey of 2,774 adults, the factors independently associated with functional gut disorders were chewing tobacco, aerated soft drinks, tea or coffee, a vegetarian diet, anxiety, and disturbed sleep. In the Delhi college study, dyspepsia and IBS tracked with fatty food, cigarettes and low physical activity. The ISG reflux task force found non-vegetarian and fried food, aerated drinks, and tea or coffee associated with weekly heartburn, with non-vegetarian food the only independent predictor on multivariate analysis.

Chilli is the one everyone blames and the data do not support in any consistent way. The ISG position statement lists spices among dietary factors associated with reflux, but the evidence is weaker than for the others, and there is no trial showing that removing chilli helps a general population of patients. If chilli reliably hurts you, stop it - but do not remove it because an article told you to.

The four that are worth a two-week test

- **Tea and coffee.** These appear repeatedly in Indian symptom surveys, in both reflux and functional disorder data. Cut to two cups and see what changes.
- **Aerated drinks.** Associated with functional disorders in two separate Indian datasets, and mechanically obvious: you are swallowing gas.
- **Fried food.** Slows stomach emptying and appears in both the reflux and dyspepsia data.
- **Tobacco, in any form.** Chewing tobacco was an independent predictor in the rural study; smoking relaxes the valve that keeps acid down. This is not a two-week test - it is a stop.

19. Meal timing beats food lists

> Three changes to when and how you eat often do more than a page of forbidden foods.

Physiology is on the side of timing. The stomach empties slowly with fat and large volumes; the valve at the top of the stomach is under pressure when you lie down on a full stomach; and the gastrocolic reflex pushes the bowel hardest in the first hour after a meal, which is why urgency after breakfast is so common.

1. **Finish the main meal three hours before lying down.** This is in NICE's guidance, the API-ISG consensus and the SAGES dyspepsia pathway, and it is free.

2. **Eat three meals and stop grazing.** Constant snacking means the stomach never reaches its between-meal clearing phase, and it keeps the bowel reflex firing all day.
3. **Halve the volume of the largest meal and add a mid-afternoon snack.** For fullness-type dyspepsia, portion size is the single biggest lever.
4. **Sit up for 30 minutes after eating.** Not a walk, not exercise - just upright.
5. **Chew properly and eat without a screen.** Swallowed air is a real contributor to belching, and hurried eating swallows more of it.
6. **Keep a 12-hour overnight gap** between the last meal and the first, if reflux is the problem. Late dinners are the commonest fixable cause in urban India.

If you do only one thing from this chapter, make it the three-hour gap before lying down. It costs nothing, it works within a week, and it is the recommendation every guideline repeats.

PART 4

The body and the brain

20. The gut-brain axis without the mysticism

> The gut and the brain talk in both directions, and in functional disorders the conversation is the disease, not a metaphor.

The Rome Foundation renamed functional gut disorders "disorders of gut-brain interaction" for a reason. In these conditions, the nerve signalling between gut and brain is altered: the gut is more sensitive to normal stretch, the brain amplifies the signal, and stress changes both motility and perception.

The Indian data make the link concrete. In the north India college study, anxiety and insomnia were independent predictors of a functional gut disorder, alongside female sex and junk food. In a 2025 study of 100 IBS patients meeting Rome IV criteria at a Delhi tertiary centre, 75% had at least one psychiatric comorbidity: generalised anxiety disorder in 44%, major depression in 38%, somatisation in 23%.

Read that carefully. It does not mean your gut symptoms are imaginary, and it does not mean the anxiety caused the gut problem. It means the two travel together, and treating only one of them leaves half the problem in place.

What this changes

It changes what counts as treatment. Gut-directed cognitive behavioural therapy and gut-directed hypnotherapy have trial evidence in IBS and are recommended in UK guidance. Low-dose tricyclic antidepressants are used not as antidepressants but as gut nerve modulators - amitriptyline at 10 mg, titrating to 30 to 50 mg, is the regimen in the British Society of Gastroenterology's dyspepsia guideline. These are prescription decisions, not self-treatment, but knowing they exist matters: many patients spend years cycling through antacids without ever being offered them.

21. Sleep and stress: the two levers nobody uses

> Disturbed sleep was an independent predictor of gut disorders in Indian data. It is also the easiest thing on this list to change.

Sleep affects gut motility, pain threshold and the composition of the gut bacteria. In the rural Uttar Pradesh study, disturbed sleep was associated with functional gut disorders on univariate analysis and formed part of the anxiety cluster that survived multivariate analysis. In the college study, insomnia was an independent predictor.

The practical target is unglamorous: seven to eight hours, a consistent wake time, no screens for the last 30 minutes, and no heavy meal within three hours of bed. If reflux wakes you at night, raising the head of the bed by 15 to 20 cm - by putting blocks under the frame, not by stacking pillows - is a mechanical fix that works.

Stress, specifically

Slow breathing at about six breaths a minute for five minutes, twice a day, is the cheapest intervention with any plausible mechanism: it increases vagal tone, which changes gut motility. The evidence for it in

IBS specifically is modest and comes from small trials. It is also free, harmless, and takes five minutes, which is a reasonable trade.

What to avoid is the trap of believing that if you were less stressed you would not have symptoms. That framing turns a medical problem into a character flaw, and it stops people asking for treatment that works.

22. Movement: what exercise changes in the gut

> Regular aerobic activity shortens gut transit time and is recommended in every major dyspepsia and constipation guideline. The effect is real but modest.

Physical activity is one of the few interventions that appears in guidance for reflux, dyspepsia, constipation and IBS at once. The mechanism is partly mechanical - movement speeds colonic transit - and partly metabolic. Low physical activity was a correlate of both dyspepsia and IBS in the Delhi college study, and of functional disorders in the rural survey.

The prescription that matches the evidence is 150 minutes a week of moderate activity, which is 30 minutes on five days, and it does not need to be a gym. A brisk 20-minute walk after dinner does two things at once: it takes you off the sofa after eating, and it stimulates the colon reflex about 30 to 60 minutes later, which is useful if constipation is the problem.

Two caveats. Vigorous exercise on a full stomach worsens reflux - leave two hours. And in IBS with diarrhoea, running can trigger urgency; walking and cycling are better tolerated.

23. Yoga and pranayama: honest verdict

> Useful for stress and for the abdominal wall. Not a substitute for testing and treating H. pylori, and not proven to change the disease.

Yoga is recommended in Indian clinical practice and is widely used. The trial evidence in functional gut disorders is small, often unblinded, and inconsistent in what it measures. That does not make it worthless; it makes it unproven for hard outcomes.

Where it plausibly helps: the abdominal wall. In functional bloating, visible distension often comes from a diaphragm and abdominal muscle reflex that pushes the wall outward rather than from excess gas. Techniques that teach diaphragmatic breathing and gentle core engagement address that mechanism directly. That is a real, physical explanation, not a spiritual one.

What it cannot do: find an ulcer, eradicate H. pylori, or replace a colonoscopy in someone with rectal bleeding. Chapter 3 is not optional because you do yoga.

A reasonable programme: 10 minutes of slow diaphragmatic breathing daily, a gentle supine twist sequence, and a short walk. Four weeks. If the belly girth difference between morning and evening narrows, you have a result you can measure.

PART 5
Treatment conversations

24. Antacids, H2 blockers, PPIs: what each actually does

Three different jobs. Most people use the weakest one daily and the strongest one wrongly.

Type	How fast	How long	Best used for
Antacids (gel, chewable, syrup)	Minutes	30 to 60 minutes	Occasional mild heartburn, breakthrough symptoms
H2 blockers (famotidine, ranitidine alternatives)	30 to 60 minutes	6 to 10 hours	Mild to moderate reflux, night symptoms, rescue after PPI withdrawal
Proton pump inhibitors (omeprazole, pantoprazole, rabeprazole, esomeprazole)	1 to 4 days for full effect	24 hours per dose	Frequent or severe reflux, erosive damage, ulcer, H. pylori regimens

Two facts about PPIs that most people get wrong. They must be taken 30 to 60 minutes before a meal to work properly - taken after food, they do much less. And they are not fast: full effect takes a few days, so a PPI is not the right choice for a sudden attack of heartburn at midnight.

The ISG position statement is clear on the ladder: for occasional or mild symptoms, antacids and H2 blockers are enough; PPIs are for frequent or severe symptoms. The API-ISG consensus adds that all PPIs at equivalent doses work about the same, so there is no reason to pay a premium for a brand. Prokinetics have a limited proven role in reflux.

One newer option worth knowing about. A 2025 pan-Indian endoscopy audit found vonoprazan was the most prescribed acid suppressant among the 1,701 patients studied, at 32.3%, ahead of rabeprazole at 21.3%. Vonoprazan is a potassium-competitive acid blocker, not a PPI, and it works faster and does not need meal timing. Availability and price vary widely across India. Whether it is right for you is a prescribing decision - but if your PPI is not working, it is a reasonable question to ask.

25. Coming off a PPI without the rebound

Stopping suddenly after months on a PPI produces real heartburn that is not your disease returning. Plan the exit.

The American Gastroenterological Association's 2022 clinical practice update on deprescribing PPIs set out ten points. The ones that matter to a patient are these.

1. **Know why you are on it.** Every PPI should have a documented indication and a review date. If nobody can say why you started, that is itself the answer.
2. **Not everyone should stop.** People with a history of severe erosive oesophagitis, an oesophageal ulcer, a stricture, Barrett's oesophagus, or a high risk of upper gut bleeding should generally stay on it.
3. **Twice daily is usually too much.** Most people on a morning-and-night PPI can step down to once daily.

4. **Expect rebound.** Stopping long-term PPI causes a temporary surge of acid as the stomach's acid-producing cells readjust. Symptoms can last weeks, and they do not mean the drug must be restarted.
5. **Tapering or stopping outright are both acceptable.** One trial found no difference between the two at six months.
6. **Keep a rescue option.** An H2 blocker or antacid taken as needed cut the risk of restarting a PPI by about 80% in a small trial. On-demand PPI use also works, at the cost of about four fewer pills a week and slightly worse symptom control in a Cochrane review.

The most important line in the AGA document is the one people ignore: do not stop a PPI because you read that PPIs are dangerous. The observational links to kidney disease, fractures and dementia have not been confirmed in randomised trials, and the AGA advises against using fear of those associations as the reason to withdraw. Stop because there is no indication, not because of a headline.

26. H. pylori eradication: the 14 days that matter

> The regimen is a course of antibiotics plus acid suppression. Finishing it, and confirming cure, is what separates treatment from a temporary truce.

Standard treatment is combination therapy for 10 to 14 days: an acid suppressor plus two or three antibiotics, sometimes with bismuth. The exact combination is a prescription decision and depends on local resistance patterns, which is why Indian guidelines have moved away from the old triple therapy in many settings - clarithromycin resistance has risen sharply, and the Maastricht VI consensus, which Indian practice follows, recommends susceptibility-guided or bismuth-based quadruple therapy where resistance is high.

What you control:

1. **Finish the full course**, even if you feel better on day 3. Stopping early is how resistance is made.
2. **Take it on time.** Twice-daily and four-times-daily dosing are not suggestions.
3. **Expect the side effects.** Metallic taste, dark stools with bismuth, nausea and loose stools are common and usually not a reason to stop. Vomiting, rash or severe diarrhoea is.
4. **Do not drink alcohol** on a regimen containing metronidazole or tinidazole.
5. **Ask about the cure test.** Guidelines insist on it for people at higher cancer risk or with persisting symptoms; four weeks after finishing antibiotics, with a two-week PPI gap.

If the first course fails, do not repeat the same antibiotics. A second, different regimen is needed, and that is a gastroenterologist's call.

27. IBS: what has evidence

> Five things work in IBS. Four of them are cheap. None of them is a cure, and all of them take four weeks to judge.

IBS is a pattern, not a disease with one cause, which is why no single treatment works for everyone. The realistic goal is a 30% to 50% reduction in symptoms, not zero.

Approach	What the evidence says
Soluble fibre (psyllium, 10 to 15 g a day)	Best-supported supplement. Risk ratio 1.53 for response in a 2026 meta-analysis of 30 trials
Dietary change (FODMAP shortlist, or a structured elimination and reintroduction)	Effective in Asian trials versus conventional advice; needs a planned reintroduction phase
Peppermint oil, antispasmodics	Modest benefit for pain and cramping; widely used, cheap, low risk
Low-dose neuromodulators (tricyclics)	Recommended second-line in BSG guidance for dyspepsia and used in IBS; prescription only
Gut-directed CBT or hypnotherapy	Recommended in UK guidance; addresses the brain half of the disorder

What does not work, in the trials: wheat bran (worse than placebo in the psyllium trial, with high dropout from worsening symptoms), long-term strict exclusion diets without reintroduction, and repeated antibiotic courses without a diagnosis.

Setting expectations honestly

In the Swedish trial that compared a low-FODMAP diet against traditional IBS dietary advice, both worked about equally: 50% of the low-FODMAP group and 46% of the traditional-advice group had a clinically meaningful improvement. The lesson is not that FODMAP is useless; it is that structured dietary advice of almost any sensible kind beats no advice, and the individual's response decides which one to keep.

28. Constipation: the ladder, and the trap

> Climb the ladder one step at a time. The trap is a daily stimulant laxative that stops working and takes the bowel's habit with it.

1. **Step 1: fluid, fibre and movement.** 30 g of fibre, 2 to 3 litres of fluid, 30 minutes of walking. Four weeks. Isabgol above 10 g a day if diet alone is not enough.
2. **Step 2: toilet mechanics.** Go at the same time daily, ideally 20 to 30 minutes after breakfast when the colon reflex is strongest. Feet on a low stool so the knees are above the hips. Do not strain, and do not sit for more than five minutes with a phone.
3. **Step 3: an osmotic laxative.** Polyethylene glycol or lactulose, taken daily, titrated to a soft but formed stool. These work by holding water in the bowel and are the guideline-preferred long-term option.
4. **Step 4: a stimulant, occasionally.** Senna or bisacodyl for rescue, not as a daily habit.
5. **Step 5: review.** If steps 1 to 3 fail after eight weeks, the problem may be pelvic floor coordination, not stool hardness - and that needs a different treatment, not a stronger laxative.

Why the daily stimulant habit is a trap: it works at first, then the bowel becomes less responsive, then the dose climbs, then stopping produces several days of nothing. None of that is dangerous in the way people fear, but it is a dead end. Osmotic laxatives do not have this problem and are what guidelines recommend for chronic use.

Two things to check before any of this. Is your thyroid, calcium or iron medication part of the problem? Iron tablets and some blood pressure drugs cause constipation, and the fix is a dose change, not a laxative. And is the stool actually hard, or is it soft but difficult to pass? Those are different problems.

29. Bloating: gas versus distension

> Visible swelling and trapped gas are two different complaints. The swelling is often a muscle reflex, which is why carminatives disappoint.

In a 2025 house-to-house survey in Ernakulam district, Kerala, using Rome IV criteria, functional abdominal bloating and distension was the second most common disorder at 20.8%, behind belching disorder at 22.5%. Bloating is not a minor complaint in India; it is one of the two commonest things people bring to a clinic.

The mechanism people assume - litres of gas inflating the bowel like a balloon - is mostly wrong. Studies that measure gas volume in bloated patients usually find normal amounts. What differs is how the abdominal wall responds: in distension, the diaphragm and abdominal muscles relax outward instead of holding the belly in, and the same volume of gas produces a visible change.

What to do about each

Problem	First moves
Trapped gas, wind, gurgling	Check lactose and the FODMAP shortlist; reduce aerated drinks; treat constipation if present; simethicone helps some people
Visible evening distension	Measure girth morning and night; diaphragmatic breathing; treat constipation; a two-week FODMAP shortlist trial
Belching, especially repeated	Usually swallowed air: eat slower, no carbonated drinks, no chewing gum, treat reflux if present
Bloating with pain and altered stool	This is the IBS pattern - chapter 27 applies, not this one

One thing to stop doing: buying carminative powders and digestive churans repeatedly. If a product relieves you only while you take it, and you have been taking it for a year, it is not treating anything - it is masking a pattern that has a better answer.

30. Ayurvedic and OTC gut products: how to read them

> Traditional remedies have a real place. The problem is the shelf, not the tradition.

Indian households have used hing, ajwain, jeera, saunf, triphala, and buttermilk for digestive complaints for centuries, and some of these have plausible mechanisms. Ajwain and jeera contain volatile oils that relax smooth muscle; hing is a resin with a long culinary and medicinal history; triphala is a mild laxative combination.

The regulatory reality is what you need to know. FSSAI-regulated nutraceutical and health supplement products must carry "NOT FOR MEDICINAL USE" and may not claim to treat, cure, mitigate or prevent a disease without specific approval. That label is a legal boundary, not a marketing quirk: a product sold under those regulations has not been evaluated as a medicine.

Four checks before buying

1. **Is there a licence number and a category word on the front?** "HEALTH SUPPLEMENT", "NUTRACEUTICAL", "PROBIOTIC FOOD" must appear in capitals next to the brand name.
2. **Is the ingredient list quantified?** A gram amount per serving, not just a list of herbs.

3. **Does it promise a cure?** If the pack or the ad says it cures acidity, IBS or piles, the product is breaking the rules - which tells you something about the seller.
4. **Is it a long-term daily habit?** If yes, ask whether a tested alternative exists. For constipation, that is isabgol or an osmotic laxative; for reflux, it is a PPI.

One specific caution on the popular "detox" and "cleanse" products: a 2025 Indian business press review of the category quoted clinicians warning that overusing cleanses can cause dehydration, dependency and, in severe cases, kidney or heart problems. The same review noted that the foundation of gut health remains diet, movement, stress and sleep - the same four things in chapters 11 to 23.

31. When to ask for a gastroenterologist

> Four situations justify a referral. Most people reach one of them after months of pharmacy visits they did not need.

1. **A red flag from chapter 3**, at any point, regardless of how long you have had symptoms.
2. **No improvement after a proper trial.** That means H. pylori tested and treated if positive, four to eight weeks of correctly timed acid suppression, four weeks of fibre, and a structured dietary trial. If all four are done and nothing moved, refer.
3. **Symptoms that need a diagnosis before treatment.** Blood in stool, anaemia, weight loss, a family history of gut cancer, or a first-degree relative with colorectal cancer before 50.
4. **A condition that needs a specialist drug.** Low-dose neuromodulators, gut-directed therapy, biological treatment for inflammatory bowel disease, or a second-line H. pylori regimen after failure.

How to make the appointment count: bring the seven-day diary from chapter 17, every previous report in date order, a written list of every medicine and supplement with doses, and three questions at the top of the page. Fifteen minutes goes a long way when the history is already written down.

PART 6
Special situations

32. After antibiotics, after a stomach infection

A gut that changes permanently after an infection or a course of antibiotics is a recognised pattern, not a coincidence.

Post-infectious IBS is well described: a bout of gastroenteritis, and then bowel symptoms that never quite settle. The risk rises with the severity of the initial infection and with the use of antibiotics for it. In India, where both are common, it is a frequent story in a gastroenterology clinic.

Antibiotics do something separate and more direct: they wipe out a share of the gut bacteria, and recovery is not always complete. Diversity often returns over weeks to months, but not always to the same composition.

What to do in the first month

1. **Do not add a probiotic blindly during the course.** Specific strains reduce antibiotic-associated diarrhoea, but a random sachet may not. If you want one, choose by strain code, as in chapter 16.
2. **Rehydrate properly** during any acute infection - oral rehydration solution, not just water.
3. **Reintroduce food slowly.** Plain rice, curd, khichdi, banana, boiled vegetables, for three to five days, then normal food.
4. **Watch the four-week mark.** If bowel habit is still wrong at four weeks after the infection cleared, that is worth mentioning to a doctor rather than waiting a year.
5. **Do not chase it with repeated antibiotic courses.** This is the commonest mistake, and it makes the problem worse.

Fermented foods are the most sensible long-term move, on the strength of the Stanford trial in chapter 15: they raise diversity, and diversity is what antibiotics reduce.

33. Hostel, mess, travel and night shifts

Three situations where the standard advice fails, and what to do instead.

Hostel and mess food

Mess food is high in refined grain, low in pulses relative to volume, and often served at fixed times in large quantities. The three highest-yield changes: keep a jar of roasted chana or peanuts for the 4 pm gap, take curd at every meal where it is available, and drink water through the day rather than at meals only. If the mess dal is watery, ask for a second serving rather than adding bread.

Travel

Traveller's diarrhoea is usually bacterial and usually self-limiting in two to four days. Oral rehydration is the treatment; antibiotics are for severe cases and are a doctor's decision. Carry ORS sachets, not just tablets. If you have IBS, travel is a common trigger for a flare - eat smaller meals, avoid the airline salad,

and keep your fibre routine going.

Night shifts

Shift work disrupts the gut in two ways: meal timing moves into the night, when gastric emptying and acid production are on a different schedule, and sleep loss lowers the pain threshold. In the Indian surveys, disturbed sleep was a consistent predictor of functional gut symptoms. Practical fixes: eat the main meal before the shift rather than at 2 am, keep the night meal light, avoid tea after midnight, and protect a fixed sleep block with dark curtains.

34. Pregnancy and reflux

> Reflux in pregnancy is common, mechanical, and mostly managed without drugs. Heartburn that starts in the second trimester is usually not a disease.

Two things change in pregnancy: the growing uterus presses on the stomach, and progesterone relaxes the valve at the top of it. In an Indian study of 400 pregnant women, reflux prevalence was 9.5% in the first trimester, 43.1% in the second and 54.1% in the third - the first-trimester figure being close to the background population rate.

What helps, in order: smaller and more frequent meals, no lying down for three hours after eating, raising the head of the bed, avoiding the specific foods that trigger you, and stopping eating late at night. Antacids containing aluminium or magnesium are commonly used; sodium bicarbonate is best avoided because of the sodium load. Any acid suppression in pregnancy is a discussion with your obstetrician, not a pharmacy purchase.

Two symptoms in pregnancy are not ordinary reflux and need same-day attention: vomiting blood, and persistent vomiting with weight loss or dehydration.

35. Your parents: constipation, pills, and one conversation

> In older adults, constipation is usually a drug side effect plus dehydration plus inactivity. Fix those three before adding a fourth medicine.

The medicine list is the first place to look. Iron tablets, calcium supplements, some blood pressure drugs, anticholinergics for bladder problems, opioids for pain and antihistamines all slow the bowel. A 2025 study from Kerala found a higher prevalence of functional gut disorders in people with hypertension, which is often a marker of exactly this kind of pill burden rather than a cause in itself.

The second thing is fluid. Older adults often drink little because they are managing urinary symptoms, and mild dehydration makes stool hard. The third is movement.

The one conversation worth having

Colorectal cancer in India is rising, and it is rising in younger people - chapter 36 has the numbers. For a parent over 50 who has never had a colonoscopy, and especially one with a family history, a change in bowel habit, unexplained anaemia, or blood in stool, the conversation is worth having now rather than after a crisis. Ask the question plainly: "Has anyone looked inside your colon?" If the answer is no and there is any of the above, book it.

36. Young Indians and colorectal cancer: the number that changed

> Nearly one in three colorectal cancer patients at a major Indian cancer centre was under 45. That is why "you are too young for that" is no longer a safe answer.

A study published in the Indian Journal of Surgical Oncology analysed 964 patients treated at Kidwai Memorial Institute of Oncology, Bengaluru, between 2019 and 2024. Of them, 279 - 28.8% - were under 45 at diagnosis. Since 2018, 30% of the institute's colorectal cancer cases were in patients aged 45 or younger, most presenting at stage 3 or 4. The average age at diagnosis in that series was 50, roughly a decade younger than the Western average. Younger patients were also more likely to have aggressive histological subtypes.

Nationally, GLOBOCAN 2022 estimated about 65,000 new colorectal cancer cases and around 38,000 to 41,000 deaths in India, making it the fourth most common cancer in both sexes. Incidence has been rising by roughly 2% to 3% a year for two decades, with urban registries higher than rural.

Screening is not part of India's national cancer programme, which focuses on breast, cervical and oral cancer. A survey of 1,374 people eligible for screening colonoscopy - including health professionals and their parents - found only 7.14% had actually had one. So the system will not find this for you.

The symptoms not to explain away

- A persistent change in bowel habit - looser or less frequent - lasting more than four to six weeks.
- Blood mixed with stool, or bleeding that has been labelled piles without an examination.
- Unexplained iron deficiency anaemia.
- Persistent abdominal pain, or a lump.
- Unintentional weight loss.

If you have any of these and you are under 45, the answer is not "wait and see". Ask for a digital rectal examination and a referral. A colonoscopy is unpleasant for a day; a stage 4 diagnosis at 38 is a different order of problem. CGHS's 2025 rate for a diagnostic colonoscopy is Rs. 5,100 at a non-NABH hospital and Rs. 6,000 at a NABH one - the private market is higher, but this is not an unaffordable test.

PART 7
Worksheets

37. Worksheet: 14-day symptom and stool log

> Print this page and fill it in by hand, twice a day. Photograph it before your appointment.

Score the worst symptom of each half-day from 0 (none) to 10 (worst imaginable). Use the Bristol types from chapter 38. Measure girth at the navel, standing, once before the first meal and once before bed.

Day	AM girth	Worst AM	PM girth	Worst PM	Stools	Notes
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						

Read it at day 14. Look for three things: a girth difference above 3 cm between morning and night; a symptom that follows a specific food by 6 to 24 hours on at least three days; and a symptom that appears only on working days.

38. Worksheet: Bristol stool chart

> One number, written down daily, is worth more than a paragraph of description.

Type	What it looks like	What it means
1	Separate hard lumps, like nuts	Severe constipation - slow transit, low fibre, low fluid
2	Sausage-shaped but lumpy	Mild constipation
3	Sausage with cracks on the surface	Normal
4	Smooth, soft sausage or snake	Normal - the target
5	Soft blobs with clear-cut edges	Lacking fibre, tending loose
6	Fluffy pieces with ragged edges	Mild diarrhoea
7	Watery, no solid pieces	Diarrhoea

Record the type for every bowel movement for 14 days. A pattern of 1 and 2 with fewer than three movements a week is functional constipation by the Rome IV definition in chapter 4. A pattern alternating 1 and 6 with pain is the mixed IBS pattern. Consistently 6 and 7 with urgency needs a stool test, not a diet.

39. Worksheet: the fibre counter

Add up one ordinary day. Most people are surprised by how low the total is.

Food	Household measure	Approximate fibre
Dal, cooked (toor, moong, chana)	1 katori, about 150 g	4 to 6 g
Rajma or chole, cooked	1 katori	6 to 8 g
Roti, whole wheat	1 medium	2 g
Rice, cooked white	1 katori	0.5 g
Millet roti (bajra, jowar, ragi)	1 medium	2 to 3 g
Oats, dry	40 g	4 g
Vegetable sabzi	1 katori	1 to 2 g
Salad	1 bowl, 150 g	3 g
Fruit, whole	1 medium	2 to 3 g
Nuts and seeds	1 handful, 30 g	3 g
Curd or milk	1 katori or glass	0 g
Isabgol	1 heaped teaspoon, about 5 g	4 to 5 g

Household measures are approximate; the per-100 g values in chapter 11 are the source figures. Target 30 g a day. If your total is under 15 g, add one katori of dal and one handful of nuts before buying anything.

40. Worksheet: doctor visit sheet

Fill this in before the appointment. Hand it over at the start. It changes the consultation.

Question	Your answer
What are your three worst symptoms, in your own words?	
When did each one start, and is it getting better or worse?	
What makes it better, and what makes it worse?	
Which of the nine red flags in chapter 3 apply to you?	
Every medicine and supplement, with dose and how long	
Previous tests and reports, with dates	
Family history: stomach, oesophageal or colorectal cancer, IBD, coeliac	
What you have already tried, and for how long	
Your three questions for the doctor	

Three questions worth asking, if you have none of your own: Do I need an H. pylori test, and have I been off my acid medicine long enough for it to be accurate? What is my working diagnosis, and what would change it? What should I do if this has not improved in eight weeks?

41. Worksheet: the 30-day reset

Four weeks, one change a week. This is the sequence the evidence supports, in the order that avoids making things worse.

Week	Do this	Measure this
1	Log symptoms, stool and girth daily. No other change. Get a baseline.	The 14-day log from chapter 37
2	Fix timing: main meal three hours before bed, no lying down after eating, three meals and no grazing.	Night symptoms, morning fullness
3	Fix fibre: add one katori of dal and one handful of nuts daily, plus 10 g isabgol if constipated. Increase fluids.	Bristol type, stool frequency
4	Run one food trial: the FODMAP shortlist from chapter 13, or a dairy-free week from chapter 14. One, not both.	Pain score, girth difference
After 4 weeks	Judge against week 1. If nothing moved, book an appointment with the sheet from chapter 40.	Your week 1 numbers

Do not run two food trials at once. If both help you will not know which one did it, and you will end up on a narrower diet than you need.

42. Worksheet: cost planner

Gut problems are cheap to investigate if you go in the right order, and expensive if you buy a year of supplements first.

Item	Reference or note	Your likely cost
Consultation, gastroenterology OPD	Rs. 350	
H. pylori stool antigen or breath test	Rs. 2,457 to 2,890	
Upper GI endoscopy, with or without biopsy	Rs. 3,400 to 4,000	
Colonoscopy, with or without biopsy	Rs. 5,100 to 6,000	
H. pylori eradication course, 14 days	Prescription dependent	
Isabgol, 200 g	Retail - check locally	
Probiotic, 30-day course	Retail - check the strain and CFU	
Antacid or PPI, 30 days	Retail - check locally	

CGHS rates are from the office memorandum dated 3 October 2025, effective 13 October 2025, metro tier, and are a floor for what a self-paying patient can expect. Private hospitals charge more. Use the last column to write what you were actually quoted, and compare three quotes before booking any scope.

Sources

Every figure in this playbook is traceable to one of the sources below. Accessed November 2026.

Indian clinical and dietary guidance

- ICMR-National Institute of Nutrition. Dietary Guidelines for Indians, 2024. Includes My Plate for the Day, food composition and fibre values.
- ICMR-NIN Expert Group. Nutrient Requirements for Indians: Recommended Dietary Allowances and Estimated Average Requirements, 2020 (short summary revised 2024). Dietary fibre set at 15 g per 1,000 kcal.
- Indian Society of Gastroenterology. Indian consensus on gastroesophageal reflux disease in adults: a position statement. Indian Journal of Gastroenterology.
- Indian Society of Gastroenterology and Association of Physicians of India. API-ISG Consensus Guidelines for Management of Gastrooesophageal Reflux Disease, 2020.
- Indian Society of Gastroenterology Task Force. Epidemiology and symptom profile of gastroesophageal reflux in the Indian population.
- Indian Society of Gastroenterology Task Force. Epidemiological and clinical profile of irritable bowel syndrome in India.

Indian epidemiology and clinical studies

- Meta-analysis and meta-regression of GERD prevalence in the Indian population, 2021: pooled prevalence 15.6% across nine studies and 20,614 subjects.
- Meta-analysis of H. pylori prevalence in India, 2025: 52 studies, 15 states, pooled 58%.
- Pan-Indian upper GI endoscopy audit, 2025: 1,701 patients, 171 centres, April 2024 to March 2025.
- Goyal et al. Prevalence, overlap and risk factors for Rome IV functional gastrointestinal disorders among college students in northern India. Indian Journal of Gastroenterology, 2021.
- Ghoshal et al. Frequency and risk factors of functional gastrointestinal disorders in a rural Indian population. Journal of Gastroenterology and Hepatology, 2017.
- Prevalence and patterns of functional gastrointestinal disorders among adults in Ernakulam district, Kerala, 2025. Journal of Family Medicine and Primary Care.
- Self-reported food intolerances in an Indian population: need for individualization rather than a universal low-FODMAP diet, 2023.
- Frequency of lactose malabsorption among healthy southern and northern Indian populations. American Journal of Clinical Nutrition, 2010.
- Prevalence of psychiatric disorders in patients of irritable bowel syndrome fulfilling Rome IV criteria, 2025.
- Colorectal Cancer Trends in Younger Versus Older Adults, Kidwai Memorial Institute of Oncology, Indian Journal of Surgical Oncology, 2025.
- Epidemiology of colorectal cancer: a review with special emphasis on India, 2025; GLOBOCAN 2022 estimates.
- Screening for colorectal carcinoma in India: real-world scenario, pitfalls and solutions, 2025.

International guidance and trials

- Rome Foundation. Rome IV diagnostic criteria for functional gastrointestinal disorders.
- British Society of Gastroenterology. Guidelines for the evaluation and management of dyspepsia, 2022.
- American Society for Gastrointestinal Endoscopy. The role of endoscopy in dyspepsia.
- NICE. Dyspepsia and gastro-oesophageal reflux disease in adults, quality standard QS96.
- American Gastroenterological Association. Clinical Practice Update on De-Prescribing of Proton Pump Inhibitors, Gastroenterology, 2022.
- Bijkerk et al. Soluble or insoluble fibre in irritable bowel syndrome in primary care: randomised placebo controlled trial. BMJ, 2009.
- Meta-analysis of fibre supplementation in chronic constipation. American Journal of Clinical Nutrition, 2022.
- Meta-analysis of fibre supplementation in irritable bowel syndrome. Gastroenterology, 2026.
- Wastyk et al. Gut-microbiota-targeted diets modulate human immune status. Cell, 2021.
- Implementing a FODMAP diet in Asia: addressing cultural, clinical and practical challenges, 2025; and the Mumbai-Monash Indian recipe analysis it describes.
- Hewawasam et al. Dietary practices and FODMAPs in South Asia. Journal of Gastroenterology and Hepatology, 2018.

Indian regulation and pricing

- FSSAI. Food Safety and Standards (Health Supplements, Nutraceuticals, Food for Special Dietary Use, Food for Special Medical Purpose, Functional Food and Novel Food) Regulations, 2016, with directions.
- FSSAI. Food Safety and Standards (Advertising and Claims) Regulations, 2018.
- FSSAI. Food Safety and Standards (Labelling and Display) Regulations, 2020.
- Central Government Health Scheme. Office Memorandum F.No.5-16/CGHS(HQ)/HEC/2024(Part I) dated 3 October 2025, revised rate list effective 13 October 2025.

Market context

- Mint, July 2026: gut health as a consumer category in India; Mintel finding that 88% of Indians say gut health is important.
- The Financial Express, September 2025: Indian gut health market size and clinician commentary on cleanses and supplements.